A Course Based Project Report on

Screen Time and Mental Wellness Analysis

Submitted to the

Department of Computer science and engineering

in partial fulfilment of the requirements for the completion of course

Data Engineering LABORATORY(22PC2AM202)

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled "**Screen Time and Mental Wellness Analysis "** is a bonafide work done under our supervision and is being submitted by **Devasani Bharath -(23071A0575), Dhamshettty Abhishek- (23071A0576), Paruchuri Vinay Sagar-(23071A05B3), Peddaboina Vinay Kumar- (23071A05B4)** in partial fulfilment for the award of the degree of **Bachelor of Technology** in **Computer Science and engineering** , of the VNRVJIET, Hyderabad during the academic year 2025-2026.

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DECLARATION

We declare that the course based project work entitled **"Screen Time and Mental Wellness Analysis**" submitted in the Department of **CSE**, Vallurupalli Nageswara Rao Vignana Jyothi Institute of Engineering and Technology, Hyderabad, in partial fulfilment of the requirement for the award of the degree of **Bachelor of Technology in Computer science and Engineering** is a bonafide record of our own work carried out under the supervision of **Mr. Vijayakumar Chilamkurthi, Assistant Professor, Department of CSE, and Mr. Ch. Suresh Kumar Raju, Assistant Professor, Department of CSE, VNRVJIET.** Also, we declare that the matter embodied in this thesis has not been submitted by us in full or in any part thereof for the award of any degree/diploma of any other institution or university previously.

Place: Hyderabad.

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ABSTRACT

The Screen Time and Mental Wellness Analysis System is developed to analyze structured behavioral and lifestyle data to identify patterns and relationships between digital screen usage and mental wellness. With the increasing dependence on digital devices for work, education, and entertainment, excessive screen exposure has become a major factor influencing stress levels, sleep quality, productivity, and overall psychological well-being. This project applies data engineering techniques to preprocess, transform, and analyze user screen time and lifestyle data in order to generate meaningful behavioral insights.

The dataset used in this project contains multiple attributes such as age, gender, occupation, work mode, screen time duration, sleep hours, exercise frequency, stress level, productivity score, and mental wellness index. Data preprocessing plays a critical role in improving data quality by removing redundant columns, handling missing values, eliminating duplicate records, standardizing categorical data, and validating logical relationships between related features. Feature engineering techniques are applied to verify screen time consistency and prepare the dataset for analytical and predictive tasks.

The refined dataset enables the identification of patterns such as the relationship between excessive screen time and increased stress levels, reduced sleep quality, and decreased mental wellness scores. It also helps in understanding how lifestyle factors such as exercise, social interaction, and sleep influence productivity and psychological well-being. The system transforms raw behavioral data into a structured, analysis-ready format suitable for statistical analysis, visualization, and machine learning applications.

This project demonstrates the practical application of data engineering concepts such as data cleaning, transformation, feature engineering, and dataset refinement in analyzing real-world behavioral data. The proposed system provides a scalable framework that can be extended for predictive mental wellness analysis, real-time monitoring systems, and intelligent recommendation systems to promote healthier digital habits and improve overall mental well-being.

INTRODUCTION

In the modern digital era, the widespread use of smartphones, computers, and other digital devices has significantly increased daily screen exposure among individuals. While digital technology has improved productivity, communication, and access to information, excessive screen time has also been associated with negative impacts on mental health, including increased stress levels, reduced sleep quality, decreased productivity, and lower overall mental wellness. Understanding the relationship between screen usage and mental well-being has become an important area of research in behavioral analytics and data science.

Data engineering provides an effective framework to collect, preprocess, transform, and analyze large volumes of structured behavioral data to extract meaningful insights. However, raw datasets often contain inconsistencies, redundant columns, missing values, and categorical variations that limit their analytical usability. Therefore, systematic preprocessing and feature engineering are essential to convert raw data into a structured and analysis-ready format suitable for statistical analysis and machine learning.

The Screen Time and Mental Wellness Analysis System is designed to analyze structured user data containing attributes such as screen time duration, sleep hours, stress levels, productivity scores, exercise frequency, and mental wellness index. By applying data cleaning, validation, and feature engineering techniques, the system transforms raw data into a refined dataset that supports behavioral pattern analysis and predictive modeling. The system also verifies logical relationships between related features, such as total screen time and its components, to ensure data reliability.

The primary objective of this project is to demonstrate the role of data engineering in analyzing behavioral and mental wellness data. The insights obtained from this system can help researchers, healthcare professionals, and organizations understand the impact of screen usage on mental health and productivity. Additionally, the refined dataset provides a foundation for future applications such as predictive mental wellness modeling, intelligent recommendation systems, and real-time behavioral monitoring systems to promote healthier digital lifestyles.

LITERATURE SURVEY

The rapid growth of digital technology has significantly increased daily screen exposure across different age groups, leading to growing concerns about its impact on mental health and overall well-being. Researchers have conducted several studies to analyze the relationship between screen time and psychological factors such as stress, sleep quality, productivity, and mental wellness. These studies emphasize the importance of structured data analysis and preprocessing techniques in extracting meaningful behavioral insights from large datasets.

Early research focused primarily on statistical analysis to identify correlations between screen usage and mental health indicators. These studies found that excessive screen time is often associated with higher stress levels, reduced sleep duration, and decreased overall well-being. However, traditional statistical approaches had limitations in handling large-scale and complex behavioral datasets with multiple interrelated features.

With the advancement of data engineering and data mining techniques, more sophisticated methods have been introduced to analyze behavioral and lifestyle data. Data preprocessing techniques such as handling missing values, removing duplicate records, standardizing categorical attributes, and normalizing numerical features have been widely adopted to improve data quality and reliability. Feature engineering techniques are also used to derive meaningful attributes and validate relationships between related variables, enabling more accurate and comprehensive analysis.

Recent studies have highlighted the importance of integrating multiple behavioral factors such as screen time duration, sleep patterns, exercise frequency, and social interaction to evaluate mental wellness. Machine learning techniques such as regression analysis, classification, and clustering have been applied to predict mental health outcomes and identify behavioral patterns. These approaches provide deeper insights into how digital habits influence psychological well-being and productivity.

In addition, modern behavioral analytics systems utilize data engineering pipelines to transform raw data into structured, analysis-ready datasets. Tools such as Python,Pandas, and NumPy are commonly used for data cleaning, transformation, and feature extraction. These structured datasets can then be used for visualization, predictive modeling, and decision-support systems.

The proposed Screen Time and Mental Wellness Analysis System builds upon these research contributions by applying systematic data preprocessing, feature engineering, and dataset refinement techniques to behavioral data. Unlike traditional approaches, this project focuses on transforming raw screen usage and lifestyle data into a clean and structured format suitable for analysis. This enables the identification of behavioral patterns, supports predictive modeling, and provides a foundation for intelligent systems that promote healthier digital habits and improved mental wellness.

DATASET

The project uses the Screen Time vs Mental Wellness dataset, which contains behavioral, lifestyle, and psychological indicators collected from individual users. The dataset integrates digital usage metrics with health and productivity factors to enable comprehensive behavioral analytics. Each record represents a single user's daily screen exposure, lifestyle habits, stress conditions, and mental wellness score.

The dataset contains both categorical and numerical attributes that describe demographic details, device usage distribution, sleep patterns, exercise frequency, social interaction, stress levels, and productivity measures. These features support feature engineering, correlation analysis, and predictive modeling within the data engineering pipeline.

The target variable, mental_wellness_index_0_100, represents the overall psychological well-being score of a user, enabling regression-based predictive analysis. The dataset is well-structured and suitable for preprocessing, transformation, and advanced analytical tasks such as feature importance evaluation and behavioral segmentation.

Dataset Features

Feature	Type	Description
user_id	Identifier	Unique identifier assigned to each user
age	Numeric	Age of the user in years
gender	Categorical (Binary)	Gender of the user (Male/Female)
occupation	Categorical	Professional role or occupation category
work_mode	Categorical	Mode of work (Remote, Hybrid, On-site)
screen_time_hours	Numeric	Total daily screen usage duration in hours
work_screen_hours	Numeric	Screen time spent on work-related activities
leisure_screen_hours	Numeric	Screen time spent on entertainment and personal use
sleep_hours	Numeric	Average daily sleep duration in hours
exercise_minutes_per_week	Numeric	Weekly physical activity duration in minutes
social_hours_per_week	Numeric	Weekly time spent in social interactions
stress_level_0_10	Numeric (Ordinal)	Self-reported stress level on a scale of 0–10
sleep_quality_1_5	Numeric (Ordinal)	Quality of sleep rated on a scale of 1–5
productivity_0_100	Numeric	Productivity score measured on a 0–100 scale
mental_wellness_index_0_100	Numeric (Target)	Overall mental wellness score on a 0–100 scale

Initial Dataset Sample (First Five Rows)

The following table shows the structure of the dataset (first five rows):

user_id	age	gender	occupation	work_mode	screen_time_hours	work_screen_hours	leisure_screen_hours	sleep_hours	sleep_quality_1_5	stress_level_0_10	productivity_0_100
U0001	33	Female	Employed	Remote	10.79	5.44	5.35	6.63	1	9.3	44.7
U0002	28	Female	Employed	In-person	7.4	0.37	7.03	8.05	3	5.7	78
U0003	35	Female	Employed	Hybrid	9.78	1.09	8.69	6.48	1	9.1	51.8
U0004	42	Male	Employed	Hybrid	11.13	0.56	10.57	6.89	1	10	37
U0005	28	Male	Student	Remote	13.22	4.09	9.13	5.79	1	10	38.5

Initial Observations (Before Preprocessing)

Based on an initial inspection of the first five records of the ScreenTime vs Mental Wellness dataset, the following data characteristics and preprocessing considerations were identified:

1. The dataset contains a few redundant columns (Unnamed: 15, Unnamed: 16, and Perfection Level) that consist entirely of null values, indicating artifacts from data export that should be removed to avoid noise in analysis.
2. All core analytical attributes (age, gender, occupation, work_mode, screen_time_hours, sleep_hours, stress_level, productivity, and mental_wellness_index) are populated without missing entries in the sampled rows, suggesting good structural completeness of primary variables.
3. Categorical features such as gender, occupation, and work_mode are present in textual form and will require appropriate encoding (e.g., label encoding or one-hot encoding) for use in statistical modeling and machine learning algorithms.
4. Numerical attributes are measured on different scales (e.g., stress_level on a 0–10 scale, productivity and mental_wellness_index on a 0–100 scale, and sleep_quality on a 1–5 scale), necessitating normalization or standardization to ensure comparability across features.
5. The dataset includes logically related variables (e.g., total screen_time_hours and its components: work_screen_hours and leisure_screen_hours). These relationships require validation to ensure internal consistency and detect potential entry or calculation anomalies.

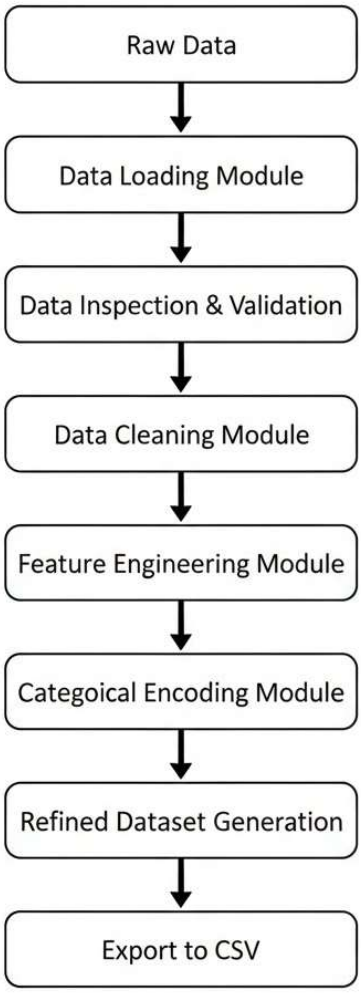
Design and Architecture

System Design Overview

The project follows a modular data engineering architecture designed to transform raw crime data into a structured and analysis-ready dataset. The architecture ensures systematic data cleaning, feature engineering, and dataset refinement using Python-based data processing tools.

The design follows a pipeline-based preprocessing model, where data flows sequentially through well-defined processing stages.

Overall Architecture Flow



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Architectural Components

Data Input Layer

Component: CSV File (ScreenTime vs MentalWellness.csv)

Purpose: Acts as the raw data source containing user screen usage and mental wellness records. The dataset initially contained:

- Redundant identifier column (user_id)
- Fully null / unnamed columns
- Mixed categorical formats
- Possible logical inconsistencies between related features

Data Loading Module

Library Used: pandas

Method Used:
pd.read_csv()
Function:
- Reads the raw CSV file into a DataFrame
- Converts tabular data into structured in-memory format
- Enables further preprocessing and transformation

Data Inspection & Validation Module
Libraries Used:
- pandas
- numpy

Methods Used:
df.info()
df.isnull().sum()
df.describe()
Purpose:
- Identify null and redundant columns
- Verify data types of each feature
- Understand statistical distribution of numerical attributes
- Validate overall dataset structure before cleaning

Data Cleaning Module
4.1 Removal of Irrelevant Columns
df.drop(columns=['user_id'])
Purpose:
- Removes non-informative identifier fields
- Drops fully null / unnamed columns
- Reduces dataset dimensionality

4.2 Duplicate Removal
df.drop_duplicates()
Purpose:
- Eliminates repeated records
- Prevents bias in statistical analysis

4.3 Missing Value Treatment
- Fully null columns removed
- Ensured completeness of remaining attributes

Purpose:
- Maintain data integrity
- Ensure no empty or unusable fields remain

4.4 Text Standardization
str.strip()
str.lower()
Purpose:
- Removes leading/trailing spaces
- Converts categorical text to lowercase
- Ensures uniform categorical formatting

Feature Engineering Module

5.1 Logical Consistency Validation

Derived feature created to verify:

screen_time_hours ≈ work_screen_hours + leisure_screen_hours

Purpose:

- Detect inconsistencies in reported screen usage
- Improve reliability of behavioral analysis

5.2 Removal of Non-Analytical Attributes

Identifier attributes such as user_id were removed as they do not contribute to analysis.

Purpose:

- Retain only meaningful analytical features
- Improve model efficiency and clarity

Categorical Encoding Module

Method Used:

pd.get_dummies() / Label Encoding

Purpose:

- Convert categorical variables (gender, occupation, work_mode) into numerical format
- Enable compatibility with statistical analysis and machine learning algorithms

Data Output Layer

Final Refined Dataset Generated:

- refined_screen_time_dataset.csv

  Method Used:

  df.to_csv()

Purpose:

- Export cleaned and structured dataset
- Enable use in visualization, statistical analysis, and machine learning workflows

Technology Stack

Component	Technology Used
Programming Language	Python
Data Processing Library	Pandas
Numerical Computation	NumPy
Encoding & ML Utilities	Scikit-learn
File Format	CSV

REFINED DATASET

After applying the preprocessing and feature engineering techniques, the crime dataset was successfully transformed into a clean, structured, and analysis-ready format.

The preprocessing pipeline resolved multiple issues present in the raw dataset, including missing values, inconsistent formatting, improper date structures, and categorical inconsistencies.

Key Improvements in the Refined Dataset

1. Removal of Irrelevant Columns
   The user_id column and fully null columns (e.g., unnamed/empty fields) were removed to eliminate unnecessary data.
2. Handling Missing and Duplicate Records
   Null columns were dropped and duplicate rows were removed to ensure a clean and consistent dataset.
3. Standardization of Categorical Features
   Categorical columns such as gender, occupation, and work_mode were standardized and encoded for analysis and modeling.
4. Validation of Numerical Ranges
   Numeric features were validated and corrected within logical bounds (e.g., stress_level: 0–10, sleep_quality: 1–5, productivity and mental_wellness_index: 0–100).
5. Consistency Check of Screen Time Features
   Total screen time was verified against work_screen_hours and leisure_screen_hours to detect inconsistencies.
6. Feature Engineering for Validation
   Derived features were created to validate logical relationships between related attributes.
7. Analysis-Ready Structured Dataset
   The final dataset is clean, consistent, and properly formatted, making it suitable for statistical analysis and machine learning tasks.

Final Refined Dataset Structure

Feature Name	Type	Description
age	Numerical	Age of the individual (in years).
gender	Categorical	Gender of the individual (encoded for analysis).
occupation	Categorical	Profession or role of the individual.
work_mode	Categorical	Mode of work (e.g., remote, hybrid, office).
screen_time_hours	Numerical	Total daily screen time in hours.
work_screen_hours	Numerical	Daily screen time spent on work-related activities.
leisure_screen_hours	Numerical	Daily screen time spent on entertainment/leisure.
sleep_hours	Numerical	Average number of sleep hours per day.
sleep_quality	Numerical	Self-reported sleep quality score (1–5 scale).
stress_level	Numerical	Stress level score on a 0–10 scale.
productivity	Numerical	Productivity score on a 0–100 scale.
mental_wellness_index	Numerical	Overall mental wellness score on a 0–100 scale.
screen_time_diff	Numerical	Difference between total screen time and sum of work and leisure screen time (validation feature).

Sample of Final Refined Dataset

age	gender	occupation	work_ mode	screen_time _hours	work_screen _hours	leisure_ screen_ hours	sleep_ hours	sleep_q uality_1 _5
33	0	0	2	10.79	5.44	5.35	6.63	1
28	0	0	1	7.4	0.37	7.03	8.05	3
35	0	0	0	9.78	1.09	8.69	6.48	1
42	1	0	0	11.13	0.56	10.57	6.89	1
28	1	3	2	13.22	4.09	9.13	5.79	1

RESEARCH ASPECTS

The primary research focus of this study is to analyze the relationship between screen time behavior and mental wellness indicators using a structured data analytics approach. With the rapid increase in digital device usage across different age groups and occupations, understanding how prolonged screen exposure influences psychological and behavioral health has become a significant area of research. This project explores how factors such as daily screen time duration, work-related screen exposure, leisure usage, sleep patterns, stress levels, and productivity collectively contribute to an individual's mental wellness index.

A key research aspect involves the transformation of raw behavioral data into a reliable analytical dataset through preprocessing techniques. Data quality plays a crucial role in producing accurate research outcomes, and therefore the study emphasizes data cleaning, removal of redundant attributes, validation of feature ranges, and consistency checks between logically dependent variables. For example, verifying whether total screen time matches the sum of work and leisure screen hours improves dataset reliability and reduces analytical errors. Such preprocessing steps ensure that the dataset reflects realistic behavioral patterns rather than noise or inconsistencies.

Another important research dimension is the application of data science tools and computational methods to behavioral health datasets. The study utilizes Python-based data processing frameworks supported by the Python Software Foundation to demonstrate scalable and reproducible data analysis workflows. By combining statistical exploration with feature engineering techniques, the research aims to identify patterns that may help predict mental wellness outcomes or detect risk factors associated with excessive digital exposure.

The research also contributes to interdisciplinary knowledge by bridging technology usage analytics with mental health awareness. The findings from such studies can support digital wellness initiatives, organizational productivity strategies, and health policy planning. Overall, the research highlights the importance of structured data engineering and analytical methodologies in extracting meaningful insights from human behavioral datasets.

CONCLUSION

This project successfully demonstrates the importance of systematic data preprocessing and feature engineering in preparing behavioral datasets for meaningful analysis. The transformation of the raw screen time and mental wellness dataset into a clean, structured, and analysis-ready format highlights the critical role of data quality in research accuracy. By removing irrelevant columns, validating numerical ranges, standardizing categorical features, and verifying logical dependencies between related variables, the dataset was refined to support reliable statistical evaluation and predictive modeling.

The analysis framework developed in this study provides valuable insights into how screen usage patterns may influence mental health indicators such as stress levels, sleep quality, productivity, and overall wellness scores. The project emphasizes that behavioral analytics requires careful consideration of data integrity, feature relationships, and preprocessing methodologies before applying advanced analytical techniques. Properly engineered datasets not only improve model performance but also enhance the credibility of research conclusions.

Another important contribution of the project is the demonstration of a scalable data processing pipeline using modern data science tools. The workflow established in this study can be adapted to other behavioral or health-related datasets, making it a reusable framework for future research initiatives. This highlights the broader applicability of data engineering practices in interdisciplinary domains involving technology usage and human well-being.

In conclusion, the project establishes a strong foundation for exploring the relationship between digital screen exposure and mental wellness outcomes. The refined dataset and preprocessing methodology enable further research, predictive modeling, and potential development of intelligent digital wellness systems. As technology continues to play an increasingly dominant role in daily life, such analytical studies will become essential for promoting balanced digital habits and improving overall quality of life.

FUTURE SCOPE

The scope for future enhancement of this project is extensive, particularly in terms of dataset expansion, methodological improvements, and real-world application integration. One major direction involves incorporating larger datasets with more diverse demographic attributes such as geographic location, socioeconomic status, educational background, and lifestyle factors. Including these additional variables would enable more comprehensive modeling of behavioral patterns and improve the generalizability of research findings across populations.

Another promising extension is the integration of real-time data collection mechanisms. Data gathered from wearable devices, smartphone usage trackers, or digital wellness applications could provide continuous monitoring of screen exposure, sleep quality, and physiological indicators such as heart rate variability. Such real-time datasets would allow dynamic analysis and early detection of behavioral risks associated with excessive screen usage. This could lead to proactive interventions rather than reactive analysis.

Future work can also explore advanced machine learning and artificial intelligence techniques to enhance predictive capabilities. Algorithms such as regression models, ensemble learning methods, clustering techniques, and deep learning architectures could be applied to predict mental wellness scores, identify user behavior segments, or detect anomalies in lifestyle patterns. Incorporating time-series analysis by adding temporal features would further enable trend forecasting and behavioral change detection over long durations.

Another significant future direction involves developing intelligent recommendation systems. These systems could provide personalized suggestions to users regarding optimal screen time limits, sleep improvement strategies, stress reduction practices, and productivity enhancement techniques. Integration with healthcare platforms or corporate wellness programs could transform the research into a practical decision-support tool for improving overall well-being.

Finally, ethical considerations and privacy-preserving analytics can be explored as part of future research. Ensuring secure handling of sensitive behavioral data while maintaining analytical accuracy will be crucial for real-world deployment. Overall, the future scope extends from academic research enhancement to practical digital health solutions.

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